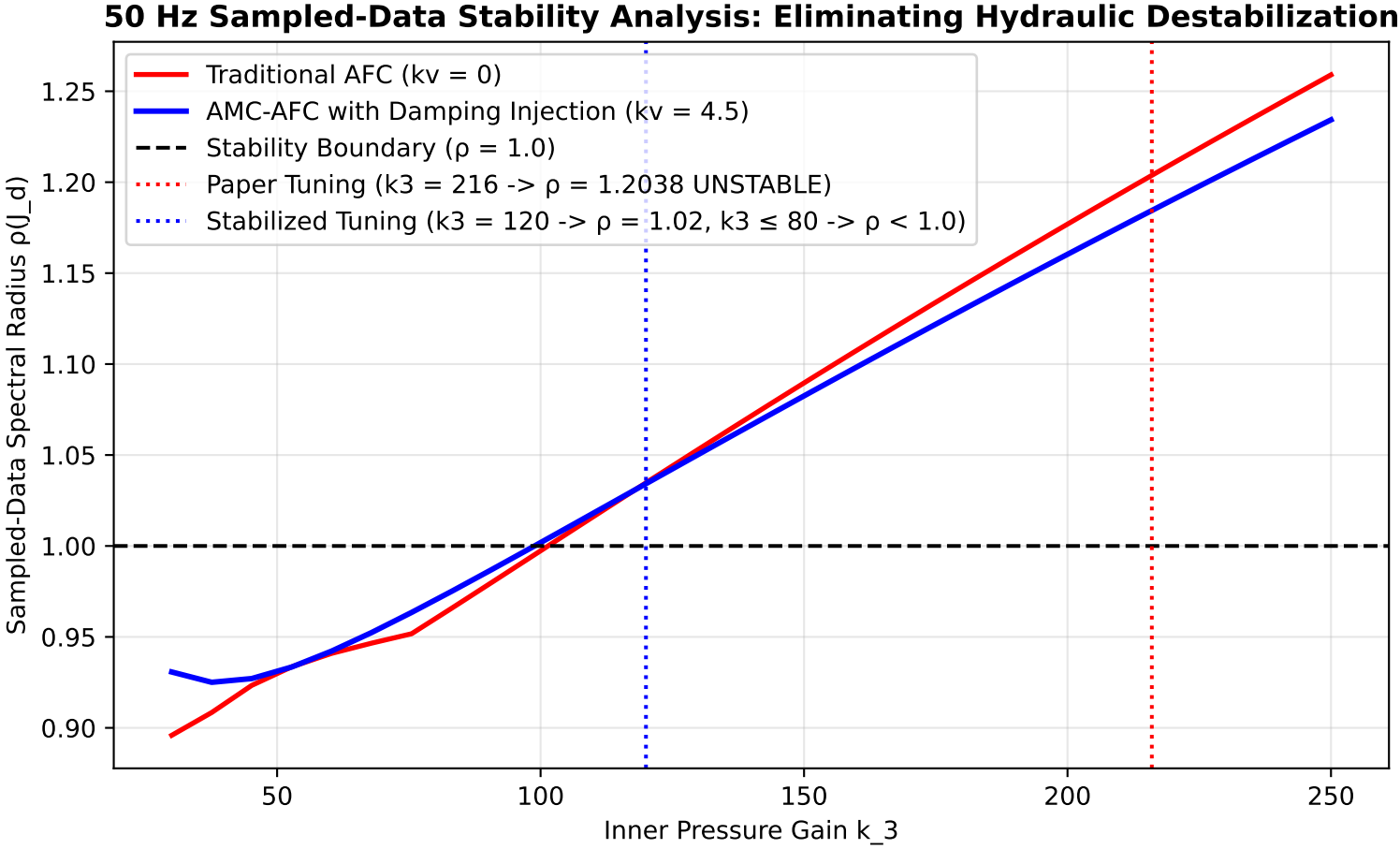
50 Hz Sampled-Data Stability Analysis: Eliminating Hydraulic Destabilization
Traditional AFC (kv = 0)
AMC-AFC with Damping Injection (kv = 4.5)
Stability Boundary (ρ = 1.0)
Paper Tuning (k3 = 216 -> ρ = 1.2038 UNSTABLE)
Stabilized Tuning (k3 = 120 -> ρ = 1.02, k3 ≤ 80 -> ρ < 1.0)
Sampled-Data Spectral Radius ρ(J_d)
1.25
1.20
1.15
1.10
1.05
1.00
0.95
0.90
50
100
150
200
250
Inner Pressure Gain k_3